Marmara University
CSE 3064– Formal Lang. & Automata Th. **Name:** _____
Midterm Homework, June 5, 2020 **Signature:** _____

Question:	1	2	3	Total
Points:	25	25	55	105
Score:				

On my honor, I have neither given nor received any unauthorized assistance on this homework. I understand that if I cheat than I will get zero grade from this online homework and I am going to be furher subject to disciplinary action by the regulations of Marmara University.

Name: _____

Signature: _____

Notes:

- Read, accept and sign the text written inside the frame above. If you cannot print this homework document, provide handwritten solutions on blank A4 sheets. In this case, write the statement inside the frame and today's date to the first solution sheet, and sign the sheet. Homework solutions without a valid signature will not be accepted.
- Write your student ID, first name and surname on top of each solution page.
- During the homework study, you may use textbooks, course slides and lecture notes.
- During the homework study, Internet access except during the homework download and submission periods is NOT allowed. You may use your computers and smart phones in offline mode.
- This homework consists of 3 questions in 4 pages.
- Any grade greater than 100 will be considered as 100.
- Please write your answers clearly and neatly to the provided spaces.
- Show all your work.
- If you make any assumptions in the solutions, please state them clearly.
- Scan all the solution pages to a single PDF file named as "FirstName_LastName.pdf" and upload the pdf file to https://ues.marmara.edu.tr/ before 13:00 on the same day.
- Unreadable homeworks will not be graded and get zero grade. Please check your pdf file before submission for readability.

1. (25 pts) Construct a deterministic finite automaton which recognizes the language described by the regular expression: $\boldsymbol{aba(aba)^*}$

2. (25 pts) Construct a non-deterministic finite automaton which recognizes the language $L = \{w \in \{a, b\}^* \mid w$ contains exactly two symbols **a** or at least three symbols **b**$\}$. e.g. strings "aba","aaabbab", "ababbbb" are in the language L, whereas "a", "abb", "ababa" are not. (Hint: You may consider designing two NFAs and construct their union.)

3. (a) (15 pts) Give the context-free grammar for the language L which contains binary numbers such that each number contains odd-number of digits, and the first, the last and the middle digits of the number are same. e.g. the strings “1”, “000”, “01000”, “10111”, “10100100001” are in the language L, whereas “100”, “1011”, “1010011” are not.

(b) (15 pts) Draw the parse tree for the string ”1011111”.

(c) (25 pts) Prove or disprove that the language L defined in part (a) above is regular.